

Tribhuvan Kumar

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OBJECTIVE

Seeking a position in an organization involved with implementation of Signaling and train control systems where expertise in the domain of Signaling engineering and project management can be leveraged for own Career growth with excellent resource management.

PROFESSIONAL PROFILE

- **An Electronics & Communication Engineer (B.E.) having around 16 years of experience in Signalling System Engineering, ideally in Metro Projects (CS03, BS02 & BS01).** It includes System Architecture, System Configuration, Technical Bid documents review, System Hardware/Software Design & Specification generation, Data Verification & Validation for Signalling Plan & Control Table, Data preparation and configuration, coordination with Independent Safety Assessor (ISA) and Commissioner of Metro Rail Safety (CMRS).
- Experience in different phase of the project such as Tendering / Bidding phase, Project Management, Contract Management, Procurement/Invoicing, Milestone Certification, Schedule Management, Interface Management, Systems Engineering, V&V, ITC, O&M in reputed and technology leader organization for signalling with Delhi Metro Rail Corporation Ltd., from client side, as Independent Engineer.
- Experience with latest Signalling technology of **DTG/CBTC (GOA4)** system including their subsystems such as Signals, Train Detection (Track Circuits/Axle Counter), Point Machine, Interlocking systems CBI/OCS, ATP/ATO/DTO/UTO system, ATS & DCS and their interface with RS/PSD/Telecom/Traction.
- In DMRC Signalling Project_CS03: acquired experience for validation of PS/GS Specification in system Configuration, Design, Architecture and RAMS of signalling and on-board system. In initial phase of ROD, have also done static and dynamic testing of 15 CBTC Trains for PAT/SAT after commissioning and provided fitness certificate for revenue service.
- A Metro Rail Signalling Engineer having 5 years of experience in Signalling Project_BS02 for Signalling design documents review for Installation/Testing & Commissioning of Train Control and wayside equipment.
- In DMRC Signalling Project_BS01, BS02 & CS03 have acquired enough experience of 10 years for Operations and Maintenance Planning, Preventive, directive and corrective maintenance of Train Control and wayside equipment.
- Proud to have worth experience of Railway Signalling System with India's Largest & reputed organization **Metro Delhi Metro Rail Corporation Ltd.** Consists of Total: 348 Kilometre current operational length passed through Phase-1, Phase-2 & Phase-3 project and O&M.
- In DMRC Operations and Maintenance; implementation of ISO-9001:14001 and OHSAS-18001 for on Line-5 & 6 of BOMBARDIER System, successfully done and SOP & Hazard/Impact/Risk Analysis bilingual report prepared for final approval.
- In DMRC Operations and Maintenance served as certified ISO auditors for implementation and sustainability of ISO quality standards in its maintenance practices.

- Familiar with BOMBARDIER (CATC-DTG & CBTC-Moving block), ALSTOM and SIEMENS signalling system practices.

CORE CAPABILITIES

- Metro Signalling Operation and Maintenance management.
- Metro Signalling interface and troubleshooting management.
- Metro Signalling Project management.
- Metro Signalling Team management.

BRIEF WORK EXPERIENCE

FROM	TO	Organization	Position	Posted at
1st Mar 2018	Till date	Delhi Metro Rail Corporation Ltd.	Sr. section Engineer	L7 BT CBTC
30 Aug 2013	27 Feb 2018	Delhi Metro Rail Corporation Ltd.	Section Engineer	L5/6 BT
05 Oct 2010	30 Aug 2013	Delhi Metro Rail Corporation Ltd.	Asst. Section Engg.	L5/6 BT
29 Sept 2005	05 Oct 2010	Delhi Metro Rail Corporation Ltd.	Junior Engineer	L1/2 Alstom
1 st Oct 1998	30 Aug2005	M/s Quasar Systems P. Ltd	Service Engineer	Delhi
1 st Oct-1998	30 Aug2005	M/s Synergy Systems P. Ltd	Service Engineer	Delhi
1 st Oct-1998	30 Aug2005	M/s Apsom InfoTech Ltd	Service Engineer	Delhi

DETAIL PROFESSIONAL EXPERIENCE

1. As SECTION INCHARGE of Bombardier CBTC Signalling and Project CS_03 of Line-7:	
Delhi Metro Rail Corporation Ltd	(1st MAR'2018 – Till date)
Designation	: Senior Section Engineer
Project	: CS03 Line-7
Type	: Bombardier CBTC Signalling System CITYFLO650
Role & Responsivities:	
● Looking after RAMS analysis of signalling and on-board system, raising and closing of system Anomaly, Punch list and modifications in design for VATC system as per PS/GS and Design documents for enhancement of system efficacy. ● Responsible for review of Technical Design & Specification documents, Signalling system circuit diagram, Signalling Plan layout, Wiring diagram & Cable layout, Maintenance/Troubleshooting Manual or any new procedure/flowchart under implementation. ● In DMRC Signalling Project_CS03, have witnessed static and dynamic testing of 15 CBTC Trains, PAT/SAT, integration test and pre-revenue service trials and provided fitness certificate for revenue service.	

- In DMRC Signalling Project_CS03, have acquired experience & exposure of on-board VATC interface with wayside-RATC, Rolling Stock, ATS, CBI, PSD, Telecom PIDS/PAS and Traction equipment.
- Proud to be the part of GOA-4 UTO based CBTC BOMBARDIER Signalling system of DMRC'S 58 KM Line-7 for 1st section opening (MKPD-DDSC): consists of 35 Trains at Mukundpur Depot. & 2nd section it's extension (TKPR-SVVR): consists of 17 Trains at Vinod Nagar Depot.
- In DMRC Operation and Maintenance, In charge of On board ATC for two depot of Line-7 (MKPD & VND) with total 55 Trains altogether equipped with **GOA-4 UTO** operations.
- Responsible for Preventive & Corrective maintenance and failure analysis of on-board VATC and Tracksides RATC with TWC & WNRA.
- Responsible for various failure analysis with CAT and CFMS BT tool and Preparation & approval of daily/weekly/monthly maintenance checklist.
- Prepared procedure of Train testing and verification of various safety parameters for Train induction into revenue service after initial commissioning.
- Prepared/reviewed procedure of retrofitting/modification of on-board hardware and uploading & testing of various VATC Operating system/Application software (GP, GA, SA, MAP, Logger etc).
- Looking after analysis, raising and closing of system Anomaly, Punch list and modifications in design for VATC system for enhancement of system efficacy.
- Involved with Training School as Instructor to newly recruited Signalling staff for Delhi Metro, Jaipur Metro, Chennai Metro & Noida Metro and evolved new training module for Bombardier DTG & CBTC System.

2. As SECTION INCHARGE of Operation & Maintenance team of Line 5 of BOMBARDIER DTG Signalling system:

Delhi Metro Rail Corporation Ltd. (30 Aug-2013 – 27 Feb-2018)

Designation : Section Engineer

Project : DMRC Line-5 & Line-6 O&M

Type : Bombardier DTG Signalling System

Role & Responsibilities:

- As leader of fifty maintenance team members; was responsible for Preventive & Corrective maintenance of Signalling & Train Control gears.
- In DMRC Operations and Maintenance, Implementation of ISO-9001 for on Line-5 & 6 of BOMBARDIER System, successfully done and SOP & Hazard/Impact/Risk Analysis bilingual report prepared for final approval.
- In DMRC Operations and Maintenance served as specialised ISO auditors for implementation and sustainability of ISO standards in its maintenance practices.
- Raising of system Anomaly and Punch list for ATC On-board & wayside equipment and its interface system.
- Developed methodology and chronological procedure for programming System & Application SW for CBI server and On-board ATP module.
- Testing of all Contractual Spare at testing Platform for Line-5 and Line 6 Signalling gears for ATC On-board & Tracksides, Ebilock950 server, OCS950, CLS, Point Machine-EEA5 & JEA72 and Power Supply System.

3. As Assistant Section Engineer for Project team of BS02 of Line 5 and Line 6 BOMBARDIER DTG Signalling System:	
Delhi Metro Rail Corporation Ltd.	(05 Oct-2010 – 30 Aug-2013)
Designation	: Assistant Section Engineer
Project	: BS02 of DMRC Line-5 & Line-6
Type	: Bombardier DTG Signalling System
Role & Responsibilities:	
● Proud member of core project_BS02 team for completion of Installation, Testing & Commissioning of Line 5 and Line 6 well before the target given for prestigious Commonwealth games-2010, was a great success milestone. ● In DMRC Signalling Project_BS02 have witnessed for Signalling system specification, architecture & design documents review, Pre/post installation test, Control table verification & validation, Testing & commissioning of wayside equipment and integrated testing of CBI and on-board ATP system. ● In DMRC Signalling Project_BS02 have done outdoor installation of Non-Trailable Point Machine-EEA5, Trailable Point Machine-JEA72, AFTC Tuning Unit and its Bond, Antenna and Earthing arrangement. ● In DMRC Signalling Project_BS02 have done Indoor installation of Power supply, CBI cubicle, OBC cubicle, ATC cubicle and Network system cubicle. ● Responsible for documents review for Signalling System Design, Specification, Architecture, ITC/ As-built Circuit Diagram and Technical manuals for approval. ● Responsible for Reviewing, Verification & Validation for final Approval of Selection Table, Control Table, Signalling & Track Plan, Cable layout Plan, Circuit Diagram for various IXL subsystems of Line5 & Line 6. ● Witness FAT, PICO, Installation, Testing & Commissioning PAT, SAT and System integration test of various indoor/Outdoor equipment of train control at Line-5 & 6. ● Responsible for Preventive, Directive & Corrective Maintenance of the equipment at Line-5 & 6 for AFTC, ATC, ATS, CLS Signals, Point Machines, CBI and UPS in Signalling & Train Control System. ● Witness in developing of Testing Platform in newly set-up Laboratory at Yamuna Bank Depot for all Signalling sub-systems for M/s Bombardier system.	

4. As Junior Engineer of Operation & Maintenance at Line 1 and Line 2 of ALSTOM Signalling system:	
Delhi Metro Rail Corporation Ltd.	(29 Sept-2005 – 05 Oct-2010)
Designation	: Junior Engineer
Project	: DMRC Line-1 & Line-2 O&M
Type	: Alstom DTG Signalling System
Role & Responsibilities:	
● Have been the part of auspicious DMRC's Operation & Maintenance team of DMRC Phase-1 when it consists of Line-1 and Line-2 altogether total length of 38 KM only. Therefore, have gone through the earl days ups-downs and challenges due to limited resources.	

- Responsible for Preventive & Corrective Maintenance of AFTC, ATC & its interface, ATS system, CLS Signals, Signal-HMAU, Point Machines, CCIP, CBI and UPS for Signalling & Train Control System.
- Witness revenue Opening of first extension of Line-2 (CS to VW section) Jahangirpuri Section of ALSTOM signalling under Phase-1 of Delhi Metro Rail Corporation Ltd.
- Involved in Installation, Testing & Commissioning and Supervised the Preventive & Breakdown Maintenance of the equipment: - SDTC, ATP & ATO, ATS, LED Signals, Point Machines, CCIP, CBI & UPS in Signalling & Train Control System.

5. Earlier Experience in other system than Railway Signalling System:

Earlier Experience: 07 Years

(1st Oct-1998 – 30 Aug-2005)

Designation : Service Engineer

Organization

M/s Quasar Systems Pvt. Ltd- An ISO 9001 Certified and Delhi Base Company- 4 year

M/s Synergy Systems Pvt. Ltd- An ISO 9001 Certified and Delhi Base Company- 2 year

- A company providing power based solution to Computer & Networking System to PSU banks, LIC and BSNL.
- Responsible for Service/Maintenance of Power supply system at site.

M/s Apsom InfoTech Ltd- An ISO 9001 Certified and Delhi Base Company- 1 Year

- A company providing Signage and Graphics automation solution in the field of Plotter and Digital printer.
- Responsible for Service/Maintenance of Signage Automation system at site.

EDUCATIONAL QUALIFICATIONS

SL.	Exam Passed	Board/University	Year of Passing	% of Marks
1.	B.E (ECE)	Jamia Millia Islamia	2021	69 %
2.	Diploma (ECE-3 YR)	Manipur University	1996	72 %
3.	12 th (HSC)	Bihar Board	1993	65 %
4.	10 th (SSC)	Bihar Board	1991	72 %

PROFESSIONAL TRAININGS

- In final year of BE Project: TPWS; prepared and successfully realised by Toy Train Model running on circular Rail guide following aspect of wayside stop signal and enforcing real time dynamic Brake. A real demonstration of ETCS Level-2 Signalling System.
- Specified onshore training for BOMBARDIER CBTC Cityflo-650 system On-board VATC specification, Design & Architecture, Operation minutes and its degraded operation scenario & it's impacts.

- Specified onshore training of one week on ANSALDO MICROLOCK, CBTC and ATS by HITACHI ANSALDO system.
- Specified onshore training of one week by Indian society for Training & Development (ISTD) for program "Training Techniques and Tools for Trainers".
- **Specified offshore training of one week on CBI Principles and design of Control Table and its verification; by Bombardier Transport at their Bangkok office in 2010.**
- Specified onshore training on ATC, OCS950 and EBISCREEN by BOMBARDIER System.
- Specified onshore training on Architecture, Functional & Troubleshooting of CBI (IPUC-EBILOCK950) by BOMBARDIER system.
- Specified onshore training of Point Machine (EEA-5 & JEA 72) by BOMBARDIER system.
- Specified onshore training of Track circuit (AFTC-TI21M) by BOMBARDIER system.
- Six Months on Job specialised Training by DMRC for Operation & maintenance of CBI based IXL & other Signalling gears and it's Troubleshooting.
- Basic signalling course of 4 weeks at Indian Railways Institute of Signal and Telecommunications Engineering (IRISET) at Ghaziabad centre, viz Signalling yard design and nomenclature of Signal, Point, Track and Routes.

ACHIEVEMENTS

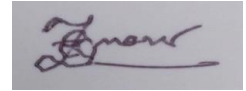
- Recipient of Positive Point for so many times during year 2011, 2013 and 2015 for outstanding contribution in restoring failure/snag of train Control system.
- Recipient of many times group cash award for outstanding performance for installation & testing of Signalling & Train Control during commissioning of Line-5 & 6.
- First time Preparation, testing and verification of six cars to four cars and vice versa done on line-5.
- Implementation of ISO-9001 for O&M on Line-5 and 6 for BOMBARDIER System successfully done and SOP & Hazard/Impact/Risk Analysis bilingual report prepared for approval.
- First time Process Preparation & Overhauling of EEA5 Point Machine and its special tools arrangement done.
- First time shifting of ESP, EKT and ATS workstation done with extension of Cables/OFC on Line-5 in DMRC.
- First time modification & retrofitting of new IPU server done due to depletion of spare at Mundka Depot.

PERSONNEL DETAILS

NAME:	TRIBHUVAN KUMAR
DATE OF BIRTH:	10 th July 1976
ORGANIZATION:	Delhi Metro Rail Corporation Limited
PRESENT POSITION:	Senior Section Engineer (In charge) for ATP-L7 (BOMBARDIER CBTC)
MARITAL STATUS:	Married
NO. OF CHILDREN:	02
TOTAL EXPERIENCE:	>23 Years
PRESENT PROJECT:	Line-7 Bombardier CBTC

PASSPORT NO.:	H8738130
LANGUAGES KNOWN:	Hindi, English
ADDITIONAL QUALITY:	Excellent team leader, hardworking & energetic member and motivator

I hereby declare that all the above statements and information are true to the best of my knowledge.



(Tribhuvan Kumar)